ELECTRIC VEHICLE MARKET INTELLIGENCE AND ANALYSIS

Project Report

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Project Tool: Tableau

Project Demo: Tableau Public

PROJECT REPORT

1. INTRODUCTION

1.1 Project Overview

The Electric Vehicle (EV) Market Intelligence project is a Tableau-based data-visualization project. It presents EV market information through worksheets and dashboards, helping users explore manufacturers, market share, vehicle categories, sales, battery technologies, driving range, customer ratings, and manufacturing geography. The submitted project is packaged as a Tableau workbook (.twbx), which includes the workbook structure and packaged data extract.

1.2 Purpose

The purpose of this project is to make EV market information easier to explore and compare. Visual dashboards help users review multiple dimensions of the dataset without having to inspect every record in a raw table. The project can support exploratory analysis, presentation of market patterns, and further research into electric vehicles.

2. IDEATION PHASE

2.1 Problem Statement

Electric-vehicle data contains many attributes, including manufacturers, vehicle models, model years, categories, battery characteristics, range, price, sales, and ratings. Looking at these attributes separately in raw records can make comparisons and patterns difficult to identify quickly. This project addresses the challenge by arranging EV data into focused visualizations and dashboards.

2.2 Empathy Map Canvas

The intended users may include students, researchers, analysts, and people interested in EV-market trends. They need to understand which manufacturers and vehicle categories appear in the data, how sales or market-share measures compare, and how battery technology relates to vehicle range. They may ask questions such as: Which manufacturers are represented? How are sales distributed? What range values are associated with battery technologies? What customer-rating information is available? The dashboard structure provides visual entry points for exploring these questions.

2.3 Brainstorming

The project organizes analysis around several themes: executive market overview, competitive intelligence and market share, global EV manufacturing, category distribution, annual sales trends, battery-technology market share, average range by battery technology, and customer-rating analysis. Together, these views provide different perspectives on the EV dataset.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Discover: the user opens the overview dashboard to understand the available information. Explore: the user selects a dashboard or worksheet that matches a question. Compare: the user reviews manufacturers, categories, sales, battery technologies, range, or ratings. Interpret: the user considers the patterns displayed and checks the measure definitions and units. Apply: the user uses the observations as a starting point for

discussion or additional research. The workbook supports exploration; it does not independently establish causal explanations.

3.2 Solution Requirement

Functional requirements include loading the EV dataset, presenting market and vehicle attributes, providing focused visualizations for sales, manufacturer distribution or share, vehicle categories, battery technologies, driving range, and customer ratings, and grouping related views into dashboards. Non-functional requirements include readable labels, clear titles, consistent measure definitions, reasonable responsiveness, and reliable access to the data source. The supplied workbook demonstrates a collection of views, but does not contain a separate formal requirements specification or measured performance targets.

3.3 Data Flow Diagram

Logical data flow: EV dataset → Tableau data source / packaged extract → worksheets and aggregations → dashboards → user interpretation. At the input stage, vehicle records provide fields for market and vehicle analysis. Tableau organizes and aggregates these fields in individual worksheets. Related worksheets are then presented through dashboards. The user reviews the resulting visualizations to identify patterns and comparisons. This is a textual logical representation based on the workbook, not an original diagram supplied with the project.

3.4 Technology Stack

Primary tool: Tableau for data visualization and dashboard creation. Project format: Tableau Packaged Workbook (.twbx). The workbook metadata includes a data source named “ev_market_test”; it also references a CSV source named “ev_market_test.csv” and a packaged Tableau extract. The analysis fields represented in the workbook relate to EV market and vehicle characteristics, including manufacturers, model names, model years, manufacturing country, market region, category, battery, range, efficiency, motor power, charging, price, sales, ratings, and use case.

4. PROJECT DESIGN

4.1 Problem Solution Fit

The solution addresses the problem by converting a multi-attribute dataset into focused charts and dashboards. Manufacturer and market-share views support competitive comparison; category views organize vehicle types; sales views support trend exploration; battery and range views support technical comparison; and customer-rating views add a customer-oriented perspective. This is a qualitative fit assessment based on the workbook structure; no formal user study or validation survey was supplied.

4.2 Proposed Solution

The proposed solution is an interactive Tableau workbook with an overview and supporting dashboards. Users can begin with a broad market view and then explore the worksheets most relevant to their questions. Dashboard names identified in the workbook include “Competitive Intelligence & Market Share Analysis,” “Executive Market Strategy & Portfolio Overview,” “Global EV Manufacturing,” and “Sustainable Mobility Research.”

4.3 Solution Architecture

Data layer: the EV market data source and packaged extract. Analysis layer: Tableau worksheets that group and aggregate relevant fields. Presentation layer: dashboards that combine related visualizations into market and technical perspectives. User layer: the person viewing the packaged workbook or the published Tableau Public visualization. The supplied project link is included in the Appendix; it should be opened once before submission to confirm that it is accessible.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

A practical sequence for the project is: (1) define the market questions; (2) inspect the dataset and identify useful fields; (3) connect and validate the data; (4) create individual worksheets; (5) combine related worksheets into dashboards; (6) refine titles, labels, layout, and formatting; (7) test the workbook with representative questions; and (8) publish the project and prepare the report. The supplied workbook demonstrates completed visualizations and dashboards. A dated timeline, task-owner list, and effort estimates were not included in the supplied file, so this sequence is a suggested planning outline rather than a verified historical schedule.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Recommended functional tests include opening the packaged workbook, confirming that the data source is available, opening each dashboard, checking worksheet titles and displayed fields, comparing selected chart values with the underlying data, and verifying the published link. Recommended performance tests include recording dashboard load times, checking interaction responsiveness, and testing on the intended device and network. No measured load times, benchmark results, or formal test logs were included in the supplied workbook; therefore, quantitative performance is not claimed as verified.

7. RESULTS

7.1 Output Screenshots

The Tableau workbook contains the following worksheets: Annual EV Sales Trend by Model, Average Range by Battery tech, Battery Technology Market Share, Customer Rating Records by EV, EV Category Performance & Distribution, Electric Vehicle Distribution by Manufacturer, Ev Market Performance at a glance, Market Share By Manufacturer. The dashboards identified in the workbook are: Competitive Intelligence & Market Share Analysis, Executive Market Strategy & Portfolio Overview, Global EV Manufacturing, Sustainable Mobility Research. The available views cover market overview, category distribution, manufacturer distribution/share, annual sales trends, battery technology, range by battery technology, customer ratings, and global EV manufacturing. Insert screenshots from Tableau here before final submission. Recommended screenshots are the executive overview, Global EV Manufacturing, and one detailed analytical dashboard. The workbook was inspected for its structure, but screenshots are not embedded in this report.

8. ADVANTAGES & DISADVANTAGES

Advantages

• Presents multiple EV-market perspectives in one Tableau workbook.

• Uses visual summaries that are easier to scan than a large raw table.

• Separates the analysis into focused worksheets and thematic dashboards.

• Covers both market measures, such as sales and manufacturer share, and vehicle attributes, such as battery technology and range.

• The packaged workbook format helps bundle the workbook with its extract for portability.

Disadvantages and Limitations

• Results depend on the accuracy, completeness, representativeness, and date coverage of the dataset.

• The workbook metadata references a CSV source path; refreshing data on another computer may require reconnecting the source.

• Market-wide conclusions should not be made until the dataset provenance and coverage are validated.

• The supplied project file does not include measured performance benchmarks, formal user-testing results, or a dated project schedule.

• Output screenshots and final submission links should be verified and added before submission.

9. CONCLUSION

The Electric Vehicle Market Intelligence project presents EV information through Tableau worksheets and dashboards focused on manufacturer share, vehicle-category distribution, sales trends, battery technologies, driving range, customer ratings, and manufacturing geography. The workbook provides a useful basis for exploratory analysis and visual comparison. To strengthen the final submission, the dataset source and measure definitions should be documented, key observations should be checked against the underlying records, screenshots should be inserted, and the published project link should be tested.

10. FUTURE SCOPE

Possible future improvements include documenting the data source and refresh process; validating records and handling missing values; adding filters for region, year, manufacturer, category, and price where the fields support them; defining key performance indicators clearly; adding time-based comparisons where suitable data exists; improving accessibility and mobile layout; testing responsiveness on different devices; incorporating newer and independently verified data; and adding concise insight summaries to dashboards. These are recommendations for future work, not features confirmed to be present in the supplied workbook.